Heewoong Noh

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RESEARCH INTEREST

AI for Materials Discovery

My research focuses on developing AI and LLM-based agentic systems to accelerate materials design and synthesis, bridging computational prediction with experimental discovery.

- **Keywords**: AI-driven materials design and synthesis, LLM agents for scientific discovery

EDUCATION

KAIST, Daejeon, South Korea

- Ph.D. in Industrial & Systems Engineering — Mar 2025 – Present
- M.S. in Industrial & Systems Engineering — Mar 2023 – Feb 2025
  - Research Interest: Machine Learning, AI4Science, LLMs for Chemistry
  - Advisor: Prof. Chanyoung Park

Korea University, Seoul, South Korea

- B.S. in Industrial and Management Engineering *GPA: 3.94/4.5* — Mar 2017 – Feb 2023
- Leave of absence for military service (Aug.2018 - Mar.2020)

PUBLICATIONS

(†: Equal contribution)

CONFERENCES

[C7] A Structured LLM Framework for Inorganic Material Synthesis Planning
Heewoong Noh, Gyoung S. Na, Namkyeong Lee, Chanyoung Park
Conference on Neural Information Processing Systems (**NeurIPS 2026**)

[C6] Learning Probabilistic Compositional Representation of Crystalline Materials
Namkyeong Lee, **Heewoong Noh**, GyoungS. Na, Jimeng Sun, Tianfan Fu, Marinka Zitnik, Chanyoung Park
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2026 AI for Science Track**)

[C5] IR-Agent: Expert-Inspired LLM Agents for Structure Elucidation from Infrared Spectra
Heewoong Noh, Namkyeong Lee, GyoungS. Na, Kibum Kim, Chanyoung Park
International Conference on Learning Representations(**ICLR 2026)**

[C4] Token-Efficient Item Representation via Images for LLM Recommender Systems
Kibum Kim, Sein Kim, Hongseok Kang, Jiwan Kim, **Heewoong Noh**, Yeonjun In, Kanghoon Yoon, Jinoh Oh, Chanyoung Park
International Conference on Learning Representations(**ICLR 2026)**

[C3] 3D Interaction Geometric Pre-training for Molecular Relational Learning
Namkyeong Lee, Yunhak Oh, **Heewoong Noh**, Gyoung S. Na, Minkai Xu, Hanchen Wang, Tianfan Fu, Chanyoung Park
Conference on Neural Information Processing Systems (**NeurIPS 2025 Spotlight**)

[C2] Retrieval-Retro: Retrieval-based Inorganic Retrosynthesis with Expert Knowledge
Heewoong Noh, Namkyeong Lee, GyoungS. Na*, Chanyoung Park*
Conference on Neural Information Processing Systems (**NeurIPS 2024**)

[C1] Density of States Prediction of Crystalline Materials via Prompt-guided Multi-Modal Transformer
Namkyeong Lee†, **Heewoong Noh†**, Sungwon Kim, Dongmin Hyun, GyoungS. Na, Chanyoung Park
Conference on Neural Information Processing Systems (**NeurIPS 2023**)

WORKSHOPS

[W3] A Structured LLM Framework for Inorganic Material Synthesis Planning
Heewoong Noh, Gyoung S. Na, Namkyeong Lee, Chanyoung Park
KDD Workshop on Agentic AI for Scientific and Societal Advances (**KDD 2026 Workshop Oral**)
Earlier version: "MSP-LLM: A Unified Large Language Model Framework for Complete Material Synthesis Planning", ICLR Workshop on AI4Mat: Accelerated Materials Design (**AI4Mat 2026**)

[W2] Stoichiometry Representation Learning with Polymorphic Crystal Structures
Namkyeong Lee, **Heewoong Noh**, GyoungS. Na, Tianfan Fu, Jimeng Sun, Chanyoung Park
NeurIPS Workshop on AI for Scientific Discovery: From Theory to Practice (**AI4Science 2023**)

[W1] Predicting Density of States via Multi-modal Transformer
Namkyeong Lee†, **HeewoongNoh†**, Sungwon Kim, Dongmin Hyun, GyoungS. Na, Chanyoung Park
ICLR Workshop on Machine Learning for Materials (**ML4Materials 2023**)

PREPRINTS

[P1] CondMSP: Condition-Informed and Resource-Aware Inorganic Materials Synthesis Planning
Heewoong Noh, Gyoung S. Na, Elton Pan, Namkyeong Lee, Chanyoung Park
arXiv preprint arXiv:2610.15603 (**2026**)
A new framework for condition- and execution-aware synthesis planning

PROJECTS

Samsung Advanced Institute of Technology (SAIT) 2026 – 2027
- Development of a Reward System for Reasoning Models in Semiconductor Materials and Devices

Korea Research Institute of Chemical Technology (KRICT)
- Geometric Deep Learning for Molecular Interactions 2024
- Retrosynthesis Analysis for Inorganic Materials 2023

AWARDS & SCHOLARSHIPS

Gold Reviewer, ICML 2026
- ICML 2026 - Recognized as a Gold Reviewer (top 25% of reviewers)

HAICon2021, Korea Institute of Information Security & Cryptology (KIISC) Nov 2021
- Awarded 6th place among 177 participating teams (AI Competition)
- Building an AI model for detecting security threats in industrial control systems (Time series anomaly detection)

Industrial Engineering Project Competition, Korean Institute of Industrial Engineers(KIIE) 2021
- Participation award in industrial engineering project competition
- Building UAM dynamic corridor algorithm

Veritas Scholarship, Korea University 2021
- Research on the detection of anomalies in the manufacturing process
  - Advisor: Prof. Sungwon Han

Certificate of Recognition, National Police Agency Oct 2019
- Awarded when serving military service as an auxiliary police

PROFESSIONAL SERVICES

Conference Reviews
- Conference on Neural Information Processing Systems(NeurIPS) 2024 – 2025
- International Conference on Machine Learning (ICML) 2025 – Present
- International Conference on Learning Representations (ICLR) 2025 – Present
- Conference on Information and Knowledge Management (CIKM ADS) 2025 – Present
- Conference on Information and Knowledge Management (CIKM Main&Short) 2025

Journal Reviews
- Transactions on Machine Learning Research(TMLR) 2026 – Present

Workshop Reviews
- AI for Accelerated Materials Design(AI4Mat)@NeurIPS 2024 – Present
- AI for Accelerated Materials Design(AI4Mat)@ICLR 2025 – Present

TALKS AND SEMINARS

Learning Probabilistic Compositional Representation of Crystalline Materials
- Lightning TALK: KDD 2026 Workshop on Reliable Scientific Foundation Models (RelSciFM) 2026

Retrieval-Retro: Retrieval-based Inorganic Retrosynthesis with Expert Knowledge
- TALK: Top Conference Session of Korea Computer Congress(KSC) 2025

Density of States Prediction of Crystalline Materials via Prompt-guided Multi-Modal Transformer
- TALK: Top Conference Session of Korea Software Congress(KSC) 2023

REFERENCES

- **Prof. Chanyoung Park**, Associate Professor, KAIST
  Email: cy.park@kaist.ac.kr

[*CV compiled on 2026-10-18*]